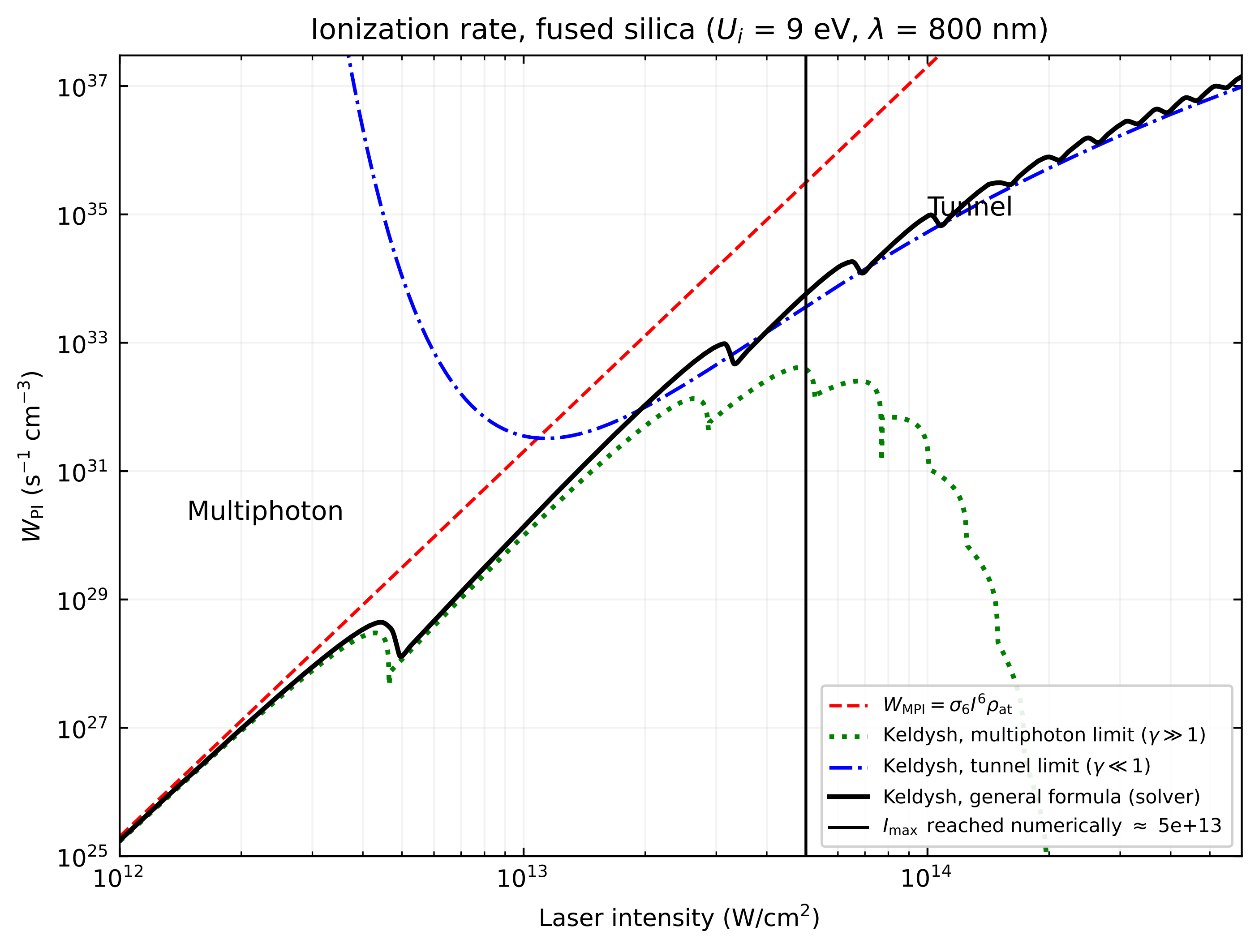


Ionization rate, fused silica (U_i = 9 eV, λ = 800 nm)
10^{37}
10^{35}
10^{33}
10^{31}
10^{29}
10^{27}
10^{25}
W_{PI} (s^{-1} cm^{-3})
10^{12}
10^{13}
10^{14}
Laser intensity (W/cm^2)
Tunnel
Multiphoton
$W_{\mathrm{MPI}} = \sigma_6 I^6 \rho_{\mathrm{at}}$
Keldysh, multiphoton limit ($\gamma \gg 1$)
Keldysh, tunnel limit ($\gamma \ll 1$)
Keldysh, general formula (solver)
I_{max} reached numerically ≈ 5e+13

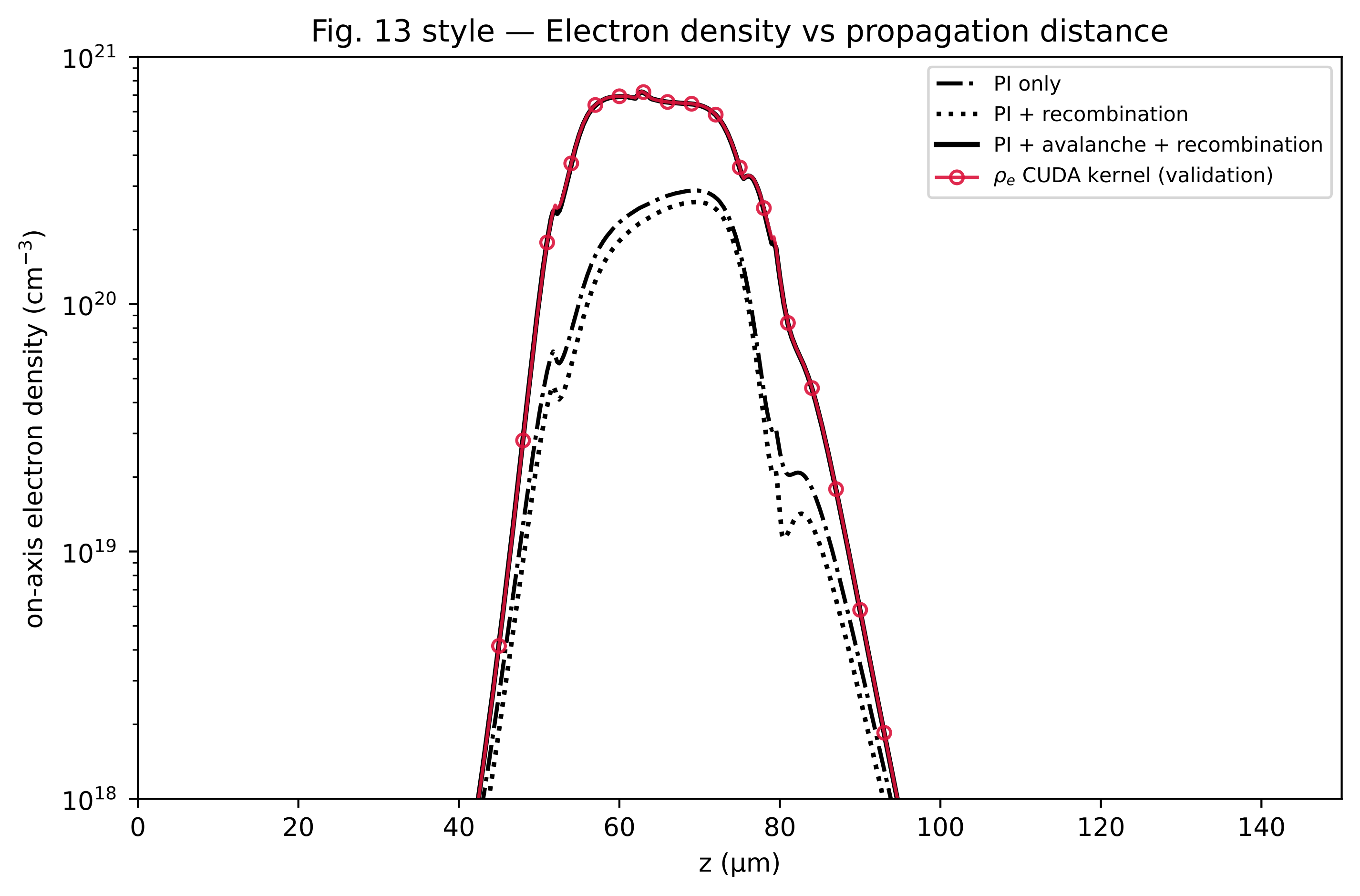
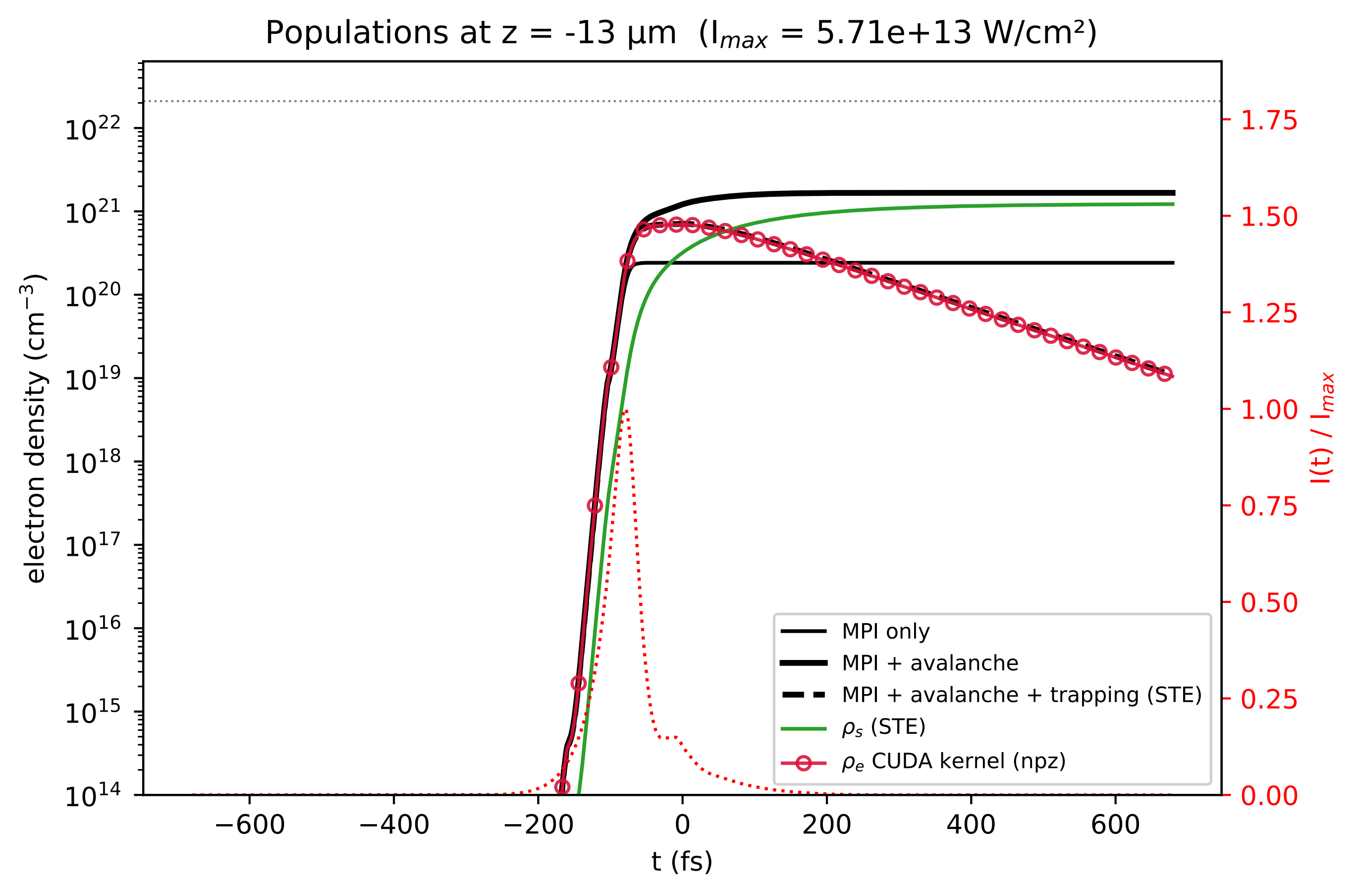


Fig. 13 style — Electron density vs propagation distance
on-axis electron density (cm^{-3})
z (μm)
10^{21}
10^{20}
10^{19}
10^{18}
0
20
40
60
80
100
120
140
PI only
PI + recombination
PI + avalanche + recombination
ρ_e CUDA kernel (validation)



Populations at z = -13 μm (I$_{max}$ = 5.71e+13 W/cm²)
electron density (cm^{-3})
10^{14}
10^{15}
10^{16}
10^{17}
10^{18}
10^{19}
10^{20}
10^{21}
10^{22}
t (fs)
−600
−400
−200
0
200
400
600
I(t) / I$_{max}$
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
MPI only
MPI + avalanche
MPI + avalanche + trapping (STE)
ρ_s (STE)
ρ_e CUDA kernel (npz)

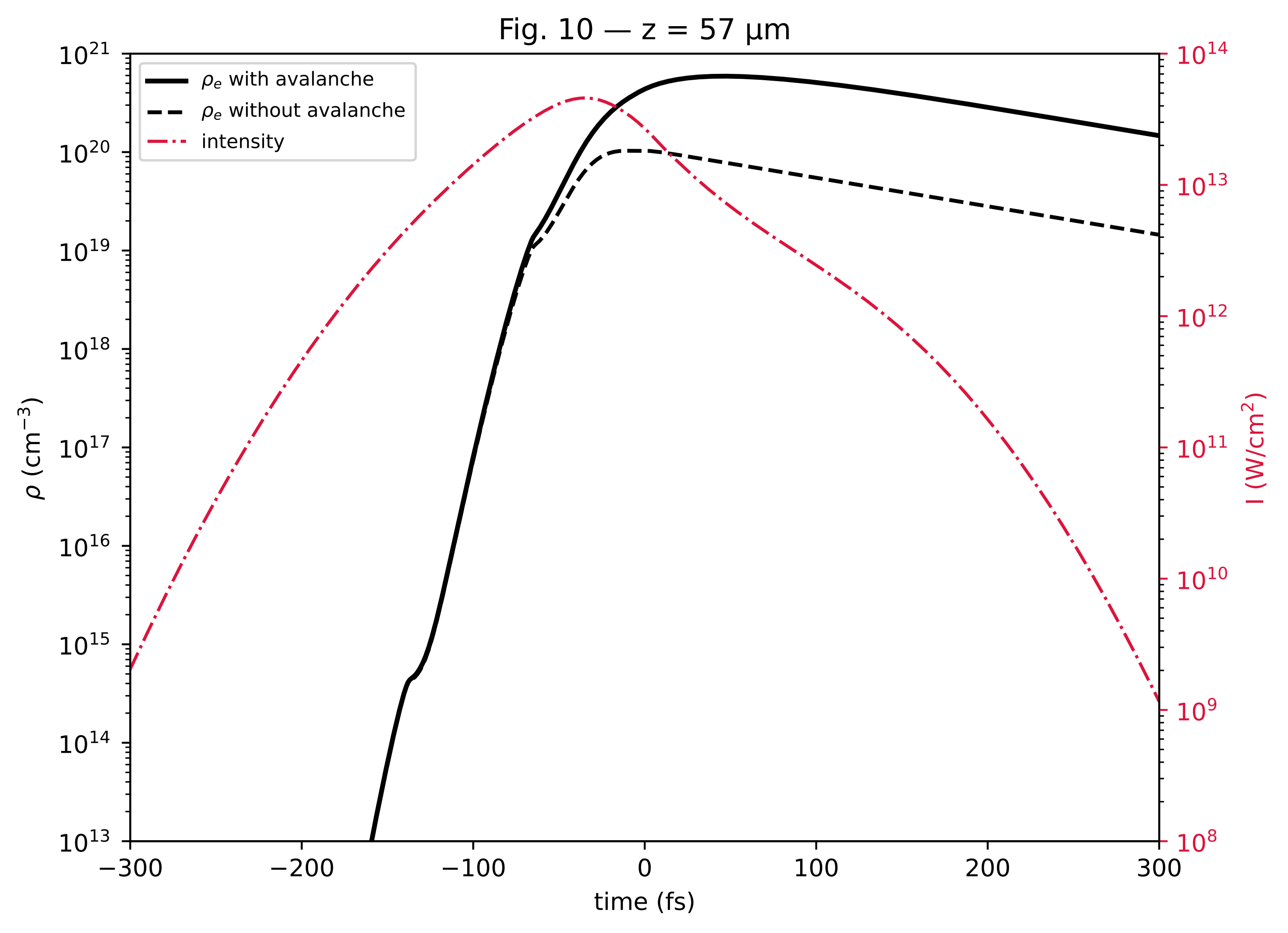
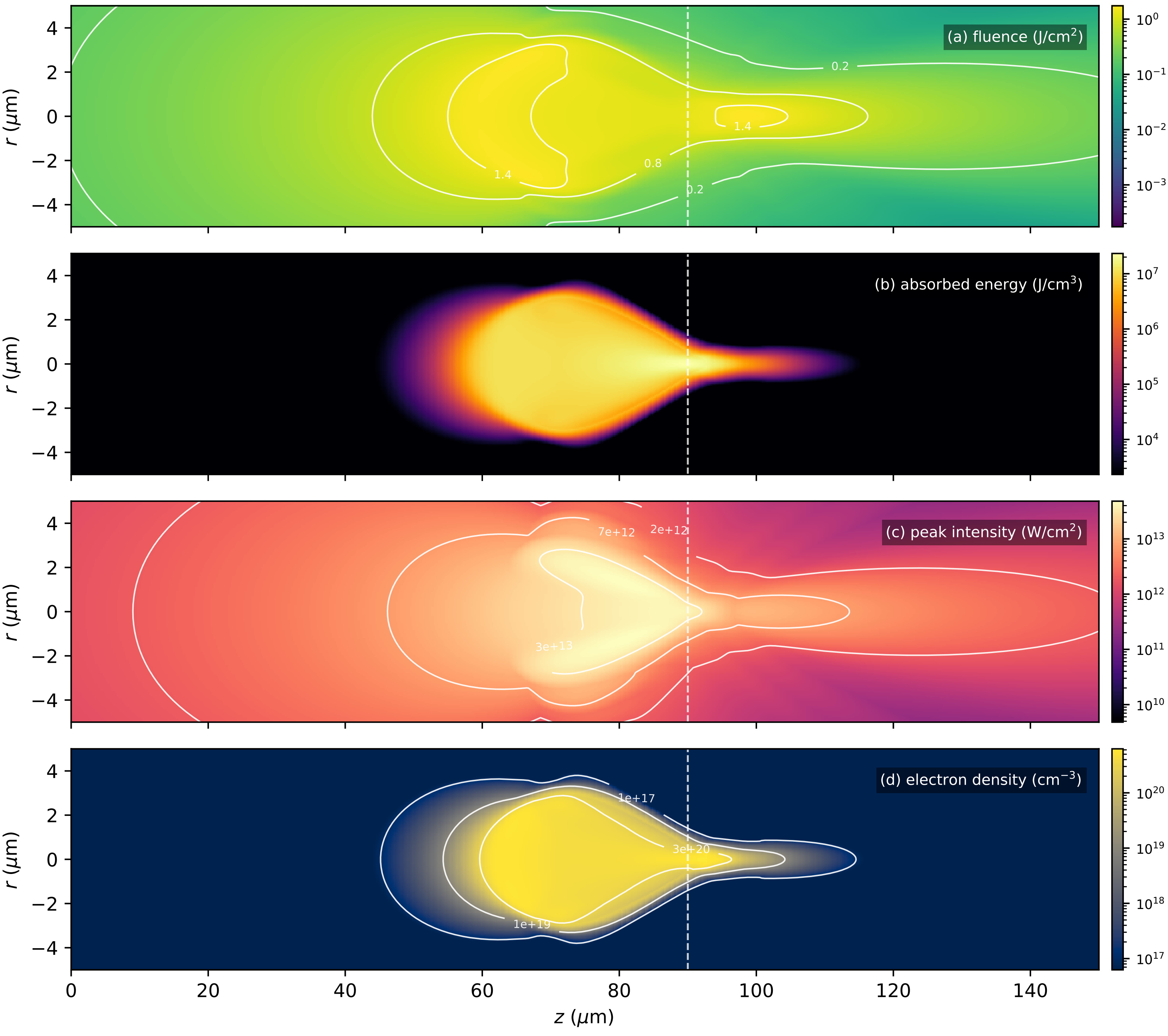


Fig. 10 — z = 57 μm
ρ_e with avalanche
ρ_e without avalanche
intensity
ρ (cm^{-3})
I (W/cm^2)
time (fs)
10^{21}
10^{20}
10^{19}
10^{18}
10^{17}
10^{16}
10^{15}
10^{14}
10^{13}
10^{14}
10^{13}
10^{12}
10^{11}
10^{10}
10^{9}
10^{8}
−300
−200
−100
0
100
200
300

Bulgakova, Stoian & Rosenfeld Fig. 11 -- fused silica, 800 nm, 120 fs, 1 μJ, w = 0.9 μm, focus at 90 μm
(dashed white line: geometric focus; white contours: published levels)



Bulgakova, Stoian & Rosenfeld Fig. 12 -- sequential energy absorption (J/cm^3), same conditions as Fig. 11
(pulse maximum at t = 0; shared colour scale across the four windows)